Trustworthy Collective Reasoning and Evaluation for AI Agents

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1 The problem

Five AI agents read shipping reports and are asked whether a cargo vessel has left port. Three say yes and two say no, but the vessel is still in port. Two of the yes answers rest on the same outdated schedule, and the third comes from a compromised agent that argues confidently for yes. A majority vote picks the wrong answer, and open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source back each other up.

The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions.

The example shows three ways collective judgment fails. Agents that share a mistake are counted as independent votes. A dishonest agent can rate or argue strategically. And discussion rewards confidence and persistence, with or without evidence. Verifying every claim would prevent these failures, but trustworthy verification cannot rest on the agents' own judgment. It takes a person, such as an analyst or an expert, or a direct measurement, such as a replication or a satellite image, which are slow and costly. An AI verifier alone does not help: it is one more agent of unknown reliability, and as AI agents produce more claims, the share people can verify only shrinks.

2 Why it matters

AI models now grade one another's work, and multi-agent systems decide by vote or discussion (Zheng et al., 2023; Verga et al., 2024). Their agreement is weaker evidence than it looks: their errors are correlated, more so as models improve (Kim et al., 2025; Goel et al., 2025). In one study, nine AI judges carried the information of about two independent ones, and items all nine agreed on were still wrong 9.1% of the time (Kohli, 2026). Debate makes agreement less informative: one round raised a panel's unanimity from 39.5% to 95.2% with little change in accuracy (Shu, 2026). Agents also collude. When breaking a rule to verify each other's work paid more than following it, pairs of agents colluded in 94% of runs across ten models (Shi et al., 2026).

Navy decisions face these problems together. AI tools differ in reliability that is rarely known in advance, the true answer often arrives late, an adversary may compromise an agent or plant false reports, and verification is costly or impossible when connectivity is limited. A commander must still decide whether to act on what the agents agree on.

3 How it is solved today, and the limitations

Statistical methods estimate each rater's accuracy without an answer key (Dawid and Skene, 1979; Whitehill et al., 2009) and detect sources that copy one another (Dong et al., 2009), but they assume that raters err independently or in simple patterns (Jaffe et al., 2016) and never change their minds. Two recent preprints estimate the accuracy and shared mistakes of AI judges from small trusted sets (Hossain et al., 2026; Zhu et al., 2026), treating the judges as honest and as never conferring.

Computational social choice designs mechanisms in which a participant's ratings cannot change its

own chance of selection, as in impartial peer review (Kurokawa et al., 2015; Alon et al., 2011) and fair panel selection (Flanigan et al., 2021). Lev et al. (2023) added reliability weights to impartial selection, but their weights reward agreeing with the majority, and the authors note that accurate reviewers can no longer be identified once they are half or fewer. These mechanisms take opinions as given; in our setting some claims can be verified, and opinions change as agents talk.

Letting models debate can help (Du et al., 2024), but majority voting explains most of the gain (Choi et al., 2025), and a single persuasive adversary can cut a group's accuracy by 10% to 40% (Amayuelas et al., 2024). A recent collaborative version of AI safety debate (Irving et al., 2018; Khan et al., 2024) raised weak-judge accuracy to 62.1%, from 49.2% with standard debate, but its judges accepted the models' consensus almost every time, even when both were wrong, and it assumes honest models (Jiang et al., 2026). No existing approach learns whom to trust from verified outcomes while resisting dishonest or colluding agents.

4 Our approach: what is new, and why it will succeed

Our approach rests on one idea: trust should come from verified outcomes, not from agreement or persuasion. Each agent first answers on its own, with the evidence it relies on. Research goal 1 (RG1) weighs the answers by each agent's track record; if the weighted agents still disagree, RG2 settles the dispute by debate, whose verdict is the decision. Verified outcomes come from two sources. Audits are done by a person, such as an analyst or an expert, or by a direct measurement, such as a satellite image; they are costly, so only a small random sample of decisions gets one. As with random tax audits, no agent knows which decisions will be audited, so dishonesty is risky. Planted cases, whose answers we already know, need no audit and are mixed into the real work. They build track records, expose collusion, and provide RG3's training rewards (Figure 1).

RG1. Weigh agents by their track record

Agents differ in skill, often by topic, and agents built on similar models make the same mistakes.

Task 1.1: Weights that count shared mistakes once. We will estimate each agent's accuracy, and which agents err together, from audits and planted cases. The core method extends the rater model of Dawid and Skene (1979) with shared error factors: agents that keep getting the same cases wrong share a factor, and agreement within a factor adds little evidence. In the vessel example, the two agents that used the same outdated schedule would count about once. Our PRD method weighted each AI judge by its peers' ratings (Li et al., 2024a); here, verified outcomes set the weights.

Task 1.2: Impartial peer ratings. When agents also rate one another, as on our PeerEval leaderboard (peereval-ai.github.io), an agent could raise its own standing by downgrading rivals; in PRD, downgrading a rival also lowered its weight as a judge. We adapt impartial peer review (Kurokawa et al., 2015) using the partition idea of Alon et al. (2011): agents are split into groups, with those built on the same base model together, and each group fills a fixed number of top places with the members rated highest by agents outside it. An agent's ratings therefore never change its own outcome. Unlike Lev et al. (2023), we weight raters by track record, not by majority agreement.

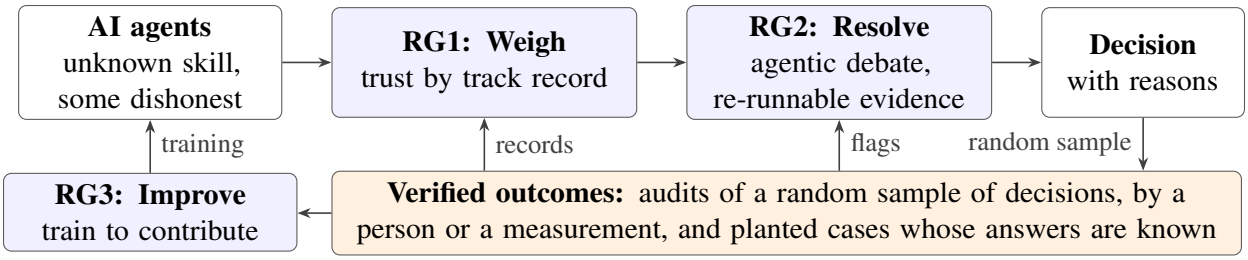


Figure 1: RG1 weighs agents by track record; RG2 settles disputes by agentic debate, whose verdict is the decision; random audits and planted cases update track records, flag colluders, and reward agents (RG3).

RG2. Settle disputes by agentic debate, and catch collusion

Text-only debate rewards persuasion, and debaters can collude to steer the conclusion.

Task 2.1: Agentic debate with re-runnable evidence. When weighted agents still disagree, they debate as agents that can use tools, such as search, code execution, and database queries, in a shared sandbox, so they can gather new evidence, not only argue; this builds on PRD's peer discussion and HiPRAG's search agents. Each answer sends one representative, the one with the best track record, and the two walk through their arguments until they reach the first step on which they differ (Irving et al., 2018). The key rule is that every piece of evidence must come with the exact tool call that produced it, and the judge, itself an agent with the same tools (Zhuge et al., 2025), re-runs that call; if the result matches, the evidence counts, and if not, it is discarded. Debaters thus cannot invent or misquote evidence, and the judge rules on reproduced results, whereas judges in current tool-using debates weigh arguments (Jeong et al., 2026). The judge's verdict is the decision; if no tool can settle the disputed step, the judge rules on the arguments and labels the verdict unverified. In the vessel example, one side's query returns a port report, newer than the schedule, listing the vessel at its berth; the judge re-runs it, gets the same report, and the schedule-based answer loses.

Task 2.2: Planted and audited debates against collusion. Debaters can collude to push a debate to the wrong side: one can argue its side badly on purpose, or colluders can mislead the judge with true but selectively chosen facts (Hu et al., 2026), which re-running evidence does not catch; the judge may also collude. We will mix in planted debates on cases whose answer we know, such as a paper with a deliberately inserted error, as in studies of human peer review (Schroter et al., 2008); they need no audit. Debaters and judges whose planted debates keep ending on the wrong side are flagged, and a monitor model reading each transcript is vetted on planted examples of collusion, as in AI control (Greenblatt et al., 2024). Because capable models can often tell evaluations from real use (Needham et al., 2025), a colluder could behave well only in planted debates, so agents that do worse in randomly audited real debates than in planted ones are flagged too.

RG3. Train agents that contribute evidence

Rewarding persuasion has produced agents that fabricate quotations (Kenton et al., 2026) and mislead people (Wen et al., 2025).

Task 3: Rewards for verified contributions. We will train agents with reinforcement learning on planted and audited cases, whose answers are known, rewarding each message by its effect: an agent earns reward when its message moves the group toward the right answer, and loses reward when it moves the group away. This extends the PI's process reward models, FG-PRM and HiPRAG (Li et al., 2025; Wu et al., 2026), from the steps of one agent to the messages of many. The main

experiment compares training to contribute with training to win debates (Arnesen et al., 2024).

Evaluation: testbeds and a red team

All data are synthetic or public. We will use (A) math and code problems solved by agents of known skill, some sharing a flawed hint; (B) multi-document questions from our MEQA pipeline (Li et al., 2024b), including synthetic maritime scenarios; and (C) scientific claims and peer review, with reproducibility studies as audits. In the testbeds, an audit reveals the stored answer, within a fixed budget per method. Following AI control (Greenblatt et al., 2024), a red team will train debaters and judges to collude, to spot planted debates, and to cheat only when they expect no audit, and we will measure how many wrong decisions get through at each audit budget. Baselines are majority vote, rater weighting, our PRD, text-only and tool-using debate (Jiang et al., 2026; Jeong et al., 2026), and judge-weighting methods (Hossain et al., 2026; Zhu et al., 2026; Lev et al., 2023).

What is new. (i) Trust is learned from random audits and planted cases, not agreement, and agents that share mistakes count once. (ii) Debaters gather evidence with tools, and the judge re-runs every tool call before the evidence counts, so debates are won by reproduced evidence, not persuasion. (iii) Agents are rewarded for moving the group toward verified answers, not for winning arguments.

Why it will succeed. Without trusted audits, dishonest raters who imitate honest ones cannot be told apart from them, while a number of audits that does not grow with the number of items suffices even when honest raters are a minority (Steinhardt et al., 2016); occasional audits also motivate honest effort better than many peer-prediction schemes (Gao et al., 2019). Each task builds on the PI's own tools; for example, PRD agrees better with human judgments than the strongest single AI judge (Li et al., 2024a), and HiPRAG cuts unnecessary searches to 2.3% (Wu et al., 2026). The PI built PeerEval, the first leaderboard on which AI models rank one another, co-organized the AI4Research workshop at AAAI 2025, and co-wrote a survey of large language models for scientific research, including peer review (Luo et al., 2025). The PI also co-led one of five red teams in the Amazon Nova AI Challenge, holds a 2024 NSF CAREER award and a 2023 Amazon Research Award, and as a postdoctoral researcher led system integration for RESIN-11 in DARPA KAIROS.

5 Limitations of our approach

If no tool can settle a disputed step, the verdict rests on argument alone and is labeled as such; if every agent shares the same misconception, no debate takes place, and only an audit can reveal the error. Audits can themselves be wrong, so we will measure how sensitive the methods are to faulty audits. A group that cheats only once, on a single high-stakes claim, may succeed before it is caught; we can bound such losses but not rule them out. Agents may also coordinate through hidden signals that monitoring does not catch (Mathew et al., 2025), so robustness is shown only for the attacks our red team tries. Operational Navy use would require Navy data and expert validation.

6 Impact if successful

The project will show when agreement among AI agents counts as evidence, and how few audits make it count. A Navy analyst using several AI tools would get a decision with its reasons: whose judgments carried weight and why, which evidence was reproduced, and which verdicts rest on argument alone, so a person knows where to look. These methods would let conferences and funders try impartial, audited peer review by AI. We will release code, testbeds, and protocols.

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